CCA 3 - Assignment

Practical Coding Assignment — Units 3, 4 and 5

Programming Language: Python
Libraries: OpenCV, NumPy, Matplotlib, scikit-image and Ultralytics YOLO
Assignment Type: Individual Practical Assignment

1. Assignment Objective

The objective of this assignment is to implement important techniques from image restoration, image segmentation, feature extraction, object recognition and motion estimation.

Each student must use their own photograph and self-recorded video to perform the assigned tasks.

Students must submit executable source code, output images, proper technical documentation, a Git repository link and a project explanation video.

2. Dataset Collection

A. Original Photograph

1. Capture one original photograph using your own mobile phone or camera.
2. Select a photograph containing a suitable object or scene with visible details.
3. Preserve the original, unedited photograph.
4. Use this photograph for the image restoration, segmentation and feature extraction tasks wherever applicable.

B. Original Video

1. Record an original video of approximately 15–30 seconds.
2. Include a moving person or object and sufficient background detail.
3. Keep the camera reasonably stable for at least part of the recording.
4. Preserve the original video without editing.
5. Use this video for object recognition, background subtraction and motion estimation.

Submit both original files with your project.

UNIT 3: IMAGE RESTORATION AND SEGMENTATION

Task 1: Image Restoration and Segmentation

Using your own photograph or a suitable frame extracted from your video, perform the following operations.

A. Image Restoration

1. Add Gaussian noise and salt-and-pepper noise to the photograph.
2. Apply median filtering and Gaussian filtering to restore the noisy images.
3. Compare the restored outputs with the original clean image.
4. Calculate PSNR and SSIM where applicable.

B. Morphological Processing and Segmentation

5. Convert a suitable image into a binary image using thresholding.
6. Apply erosion, dilation, opening and closing.
7. Apply Canny edge detection.
8. Perform watershed segmentation on a suitable image containing multiple or touching objects.

Expected output: Noisy images, restored images, binary masks, morphological results, edge maps and segmentation results.

Compare the outputs and identify which techniques work best for your selected images.

UNIT 4: FEATURE EXTRACTION

Task 2: Feature Extraction and Image Matching

Using your own photograph and a second photograph of the same object or scene:

1. Extract and visualize Histogram of Oriented Gradients (HOG) features.
2. Detect SIFT keypoints and calculate descriptors.
3. Match the SIFT features between the two images.
4. Repeat the matching experiment after rotating or resizing one image.
5. Display and compare the matching results.

Expected output: HOG visualization, SIFT keypoints, feature-matching results and a comparison of matching quality.

UNIT 5: OBJECT RECOGNITION AND MOTION ESTIMATION

Task 3: Object Recognition Using Viola–Jones and YOLO

Using your photograph or selected frames from your self-recorded video:

1. Apply a pretrained Haar cascade classifier to detect faces or another supported object.
2. Apply a pretrained YOLO model to detect supported objects.
3. Display bounding boxes, class labels and confidence scores where applicable.
4. Compare the detection results.
5. Identify at least one case in which a detector misses an object or produces an incorrect detection, if encountered.

Expected output: Detection images or video frames and a comparison of the two approaches.

Task 4: Video Motion Estimation

Using your self-recorded video:

1. Extract consecutive frames from the video.
2. Apply frame differencing to identify moving regions.
3. Apply Gaussian Mixture Model (GMM)-based background subtraction.
4. Apply Lucas–Kanade optical flow to estimate motion between frames.
5. Display the foreground masks and motion vectors.
6. Compare the results of frame differencing, background subtraction and optical flow.

Expected output: Selected video frames, foreground masks, motion visualizations and a comparison table.

3. Technical Documentation Requirements

Students must prepare basic documentation containing:

1. Input: Original photograph, original video, and details of the input data used.
2. Source Code: Code used to implement each task.
3. Output: All relevant output images, processed video frames, and results.

4. Explanation: A brief explanation of each task, the technique used, and the output obtained.
5. Comparison: Comparison of results wherever multiple techniques are applied.
6. Conclusion: A brief summary of the work performed and the results obtained.

All input and output images must be included with proper captions and brief explanations.

4. Git Repository Requirements

Every student must create a Git repository and push the complete source code to it.

The repository must contain:

- Python source files or Jupyter Notebooks.
- A README file explaining the assignment and execution steps.
- A requirements file listing the required Python packages.
- Relevant example outputs or selected result images.
- Instructions for obtaining and organizing the input dataset.

Large raw videos should be uploaded to Google Drive rather than unnecessarily committed to Git.

The Git repository URL must be clearly mentioned in the main technical report.

You need to give me access in any places it required, rahul.inchal.inchal24@gmail.com and rahul.inchal.vf@mitwpu.edu.in

5. Project Explanation Video

You must record a video of approximately 5–8 minutes explaining their completed assignment.

The video must include:

1. Introduction to the assignment and its objectives.
2. Description of the original photograph and video.
3. Explanation of the algorithms implemented.
4. Demonstration of the source code.
5. Demonstration and explanation of important input and output results.
6. Comparison of the selected techniques.

Students must explain the work in their own words. A video containing only slides or a silent screen recording will not be sufficient.

Upload the video to the Google Drive location provided by me and include its link in the main technical report.

6. Final Submission Requirements

Every student must submit the following:

1. Original photograph.
2. Original self-recorded video.
3. Complete source code pushed to Git.
4. Git repository URL mentioned in the main technical report.
5. Complete technical documentation with input images, intermediate outputs, final outputs and explanations.
6. Comparison tables and relevant performance measurements.
7. Project explanation video.
8. Any additional files required to execute the code.

Submission Locations

Google Drive: Upload all relevant project materials to the Google Drive link provided by me.

Git repository: Push the source code and related project files to Git. Mention the repository URL in the main technical report.

Main technical report: Include the Git repository URL and project explanation video link.

Before submission, verify that all required files are uploaded, the repository is accessible and the code can be executed.